Ch 4.2

0. a) $(I-P) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ ($\in C(P)$, $\in C(I-P)$)

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = (I-P)^2$$

$P^T \neq P$ so e not $\perp$ to projection e.

② $P^2 = P$, $P^T = P$

$(Pv)^T (I-P) w = v^T P (I-P) w = v^T (P-P^2) w$ ($P-P^2 \to 0$)

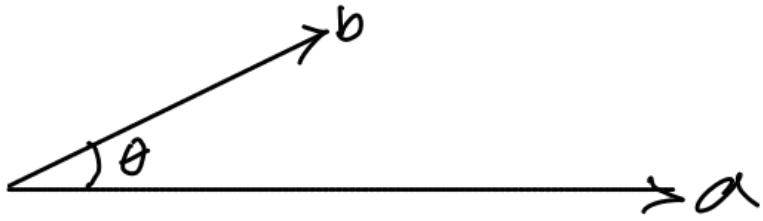
$= 0$

① proj $p = a \dfrac{a^T b}{a^T a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \dfrac{5}{3} = \begin{bmatrix} 5/3 \\ 5/3 \\ 5/3 \end{bmatrix}$

$e = b - p = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 5/3 \\ 5/3 \\ 5/3 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$ $\quad -2/3 + 1/3 + 1/3 = 0$

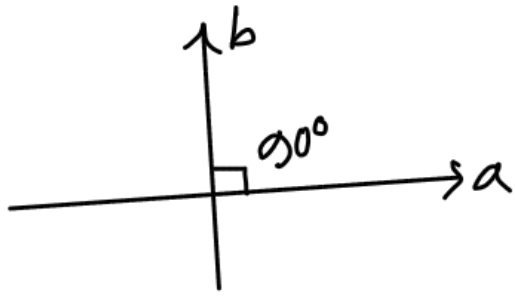
② a) $b = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$ $\quad a = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\dfrac{a^T b}{a^T a} = \dfrac{\cos\theta}{1}$ $\quad \therefore p = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos\theta = (\cos\theta, 0)$



⑥ $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $a = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$p = (0, 0)$



③ $P = \dfrac{aa^T}{a^Ta} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$P_1 b = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$

④ $P_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ → $P_1 P_2 \neq 0$, $(P_1 + P_2)^2 \neq P_1 + P_2$

Subspace (ind col)

↓

it project any vector onto it.

⑤ $a_1 \perp a_2$

⑦

21:03 Fri, 4 Apr 73%
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In linear algebra, if you have several mutually orthogonal subspaces whose direct sum equals the entire space, then the projection matrices onto these subspaces add up to the projection matrix onto the whole space. Since the projection onto the whole space is just the identity matrix, we have

$$P_1 + P_2 + P_3 = I.$$

Key Points to Include:

1. Projection Matrices:
   A projection matrix P projects any vector onto a specific subspace. It has the property that $P^2 = P$.
2. Orthogonal Subspaces:
   If subspaces are orthogonal, their projections do not overlap. That is, if v is any vector, its projection onto one subspace does not affect its projection onto another.
3. Decomposition of Vectors:
   When the subspaces are mutually orthogonal and their sum spans the entire space (e.g., $\mathbb{R}^3$ in this problem), every vector v can be uniquely decomposed as:

$$v = P_1 v + P_2 v + P_3 v.$$

$$(P_1 + P_2 + P_3)v = v,$$

$$P_1 + P_2 + P_3 = I.$$

1. Application to the Given Matrices:
   In the example, each P_i is given as a 3×3 matrix:

$$P_1 = \frac{1}{9}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{bmatrix}, \quad P_2 = \frac{1}{9}\begin{bmatrix} 4 & 4 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 1 \end{bmatrix}, \quad P_3 = \frac{1}{9}\begin{bmatrix} 4 & -2 & 4 \\ -2 & 1 & -2 \\ 4 & -2 & 4 \end{bmatrix}.$$

$$P_1 + P_2 + P_3 = I.$$

Conclusion:
This result shows that the entire space is covered by these orthogonal projections. In an exam, you should explain that the identity matrix represents the fact that every vector is fully reconstructed by its components along the orthogonal directions defined by the projections.

Ask anything

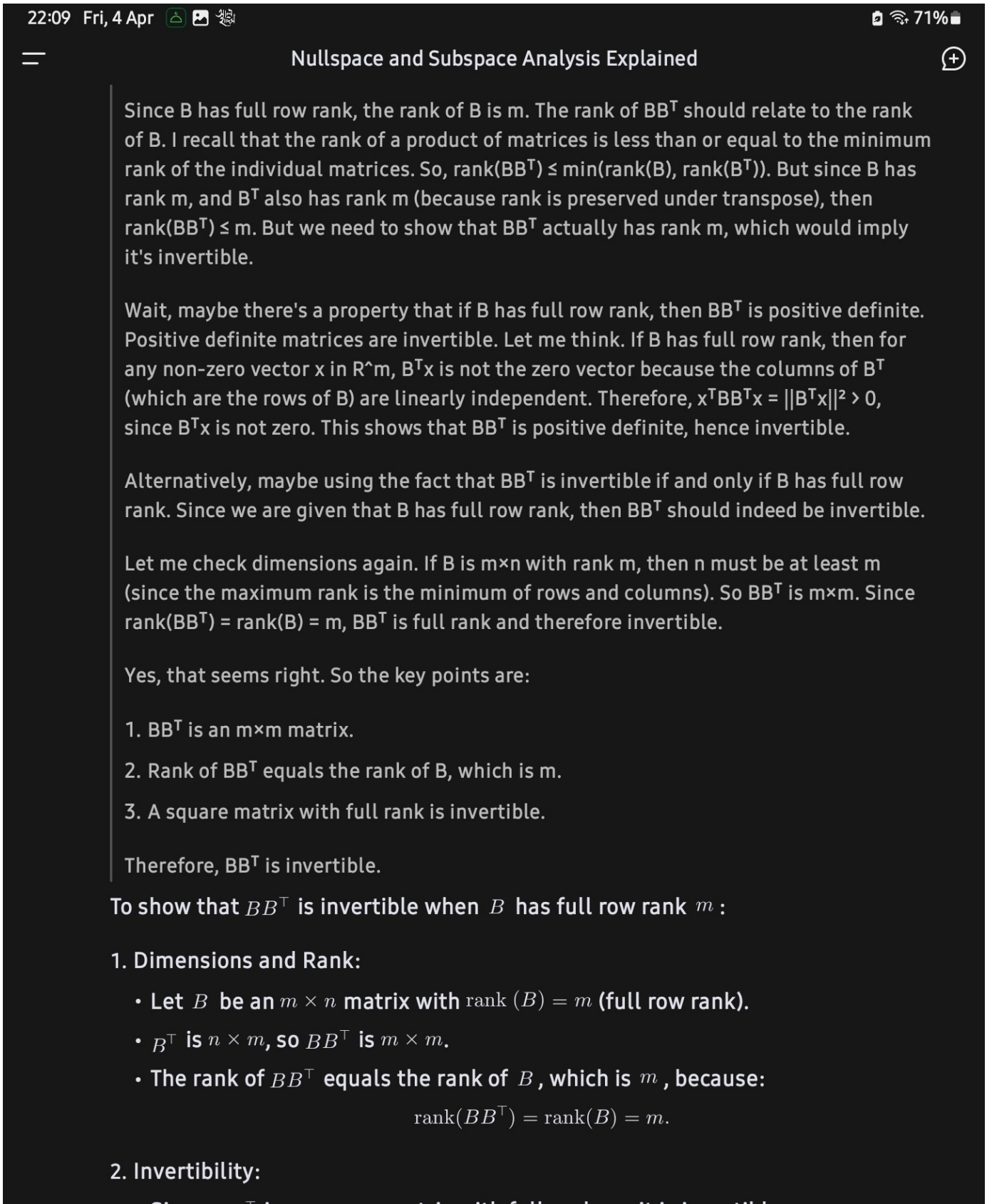
(30) B rank = m full row rank

so nullspace rank = 0

BB^T → ind col

↓ ind row

=



- Since $BB^\top$ is an $m \times m$ matrix with full rank m, it is invertible.
- Additionally, $BB^\top$ is positive definite: for any non-zero $\mathbf{x} \in \mathbb{R}^m$,

$$\mathbf{x}^\top BB^\top \mathbf{x} = \|B^\top \mathbf{x}\|^2 > 0,$$

as $B^\top \mathbf{x} \neq 0$ (due to B having full row rank).

Final Answer

New chat

Message DeepSeek

DeepThink (R1) Search

22:10 Fri, 4 Apr 71%

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Definition of Positive Definite Matrices

A symmetric matrix A (i.e., $A^\top = A$) is positive definite if it satisfies the following condition for all nonzero vectors $\mathbf{x}$:

$$\mathbf{x}^\top A\mathbf{x} > 0 \quad \text{for all } \mathbf{x} \neq 0.$$

This means that when any nonzero vector is multiplied by A in this quadratic form, the result is always strictly positive.

Key Properties of Positive Definite Matrices

1. All Eigenvalues Are Positive
   - If A is positive definite, then all eigenvalues λ of A satisfy $\lambda > 0$.
   - This ensures that A is invertible (nonsingular).
2. Determinants Are Positive
   - The determinant of a positive definite matrix is always positive.
3. Leading Principal Minors Are Positive
   - If A is an $n \times n$ matrix, all its leading principal minors (determinants of its upper-left submatrices) must be positive.
4. Cholesky Decomposition Exists
   - If A is positive definite, it can be factored as:

$$A = LL^\top,$$

Geometric Interpretation

A positive definite matrix defines an ellipsoid in geometry. It generalizes the idea of squaring a norm:

$$\mathbf{x}^\top A\mathbf{x}$$

Ask anything

Geometric Interpretation

A positive definite matrix defines an ellipsoid in geometry. It generalizes the idea of squaring a norm:

$$\mathbf{x}^\top A\mathbf{x}$$

acts like a squared norm, which is always positive.

- If $A = I$, then $\mathbf{x}^\top A\mathbf{x} = |\mathbf{x}|^2$ is just the squared Euclidean norm.
- More generally, A transforms the norm in a way that keeps it strictly positive.

Examples

✅ Example 1: Positive Definite Matrix

$$A = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}$$

$$\mathbf{x}^\top A\mathbf{x} = 2x_1^2 - 2x_1x_2 + 2x_2^2 = (x_1 - x_2)^2 + x_1^2 + x_2^2 > 0.$$

❌ Example 2: Not Positive Definite

$$B = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

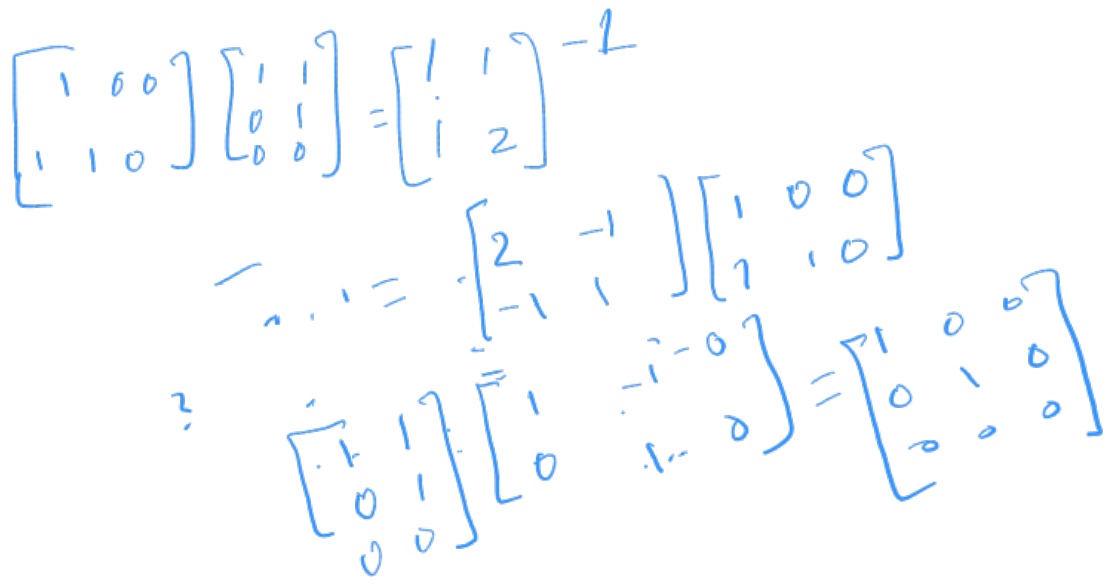
$$\mathbf{x}^\top B\mathbf{x} = 1(0) + (-1)(1) + 1(1) + (-1)(0) = -1.$$

Conclusion

✅ A positive definite matrix always produces positive quadratic forms.
✅ All eigenvalues are positive, guaranteeing invertibility.
✅ It is crucial in optimization, statistics, and physics, especially in defining energy functions and probability distributions.

$$\boxed{A \text{ is positive definite if } \mathbf{x}^\top A\mathbf{x} > 0 \text{ for all } \mathbf{x} \neq 0.}$$

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14

23:02 Fri, 4 Apr 77%

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$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 2: Projection Formula

The formula for the projection matrix onto the column space of A is:

$$P = A(A^T A)^{-1} A^T$$

$$A^T A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(A^T A)^{-1} = I$$

$$P = AIA^T = AA^T$$

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3: Projecting b

Applying P to b:

$$Pb = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$$

Final Answer

- The projection matrix P is:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Ask anything

Search Reason

15 B. E.

16 $P = A(A^TA)^{-1}A^T$, Doubling A,

$B = 2A$

$$P_B = B(B^TB)^{-1}B^T = 2A\left[(2A)^T(2A)\right]^{-1}2A^T$$

$$= 2A(4A^TA)^{-1}2A^T$$

$$= \frac{2}{4}(A^TA)^{-1}2A^T = P$$

$= \frac{1}{4} A$ (illegible)

SD & same.

is x_b same?

Least sq sol^n : $x_b = (A^TA)^{-1} A^Tb$

Now 2A,

$$x_b' = \frac{1}{4}(A^TA)^{-1} 2A^Tb = \frac{1}{2}(A^TA)^{-1} A^Tb$$

so it is halved.

(17) Given, $a_1 = (1, 2, -1)$
$a_2 = (1, 0, 1)$

Target vect $b = (2, 1, 1)$

Treating a_1, a_2 as cols of a matrix

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ -1 & 1 \end{bmatrix}$$

we wanna find $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ such that

$Ax \approx b$ or ideally $Ax = b$

$$\begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}$$

$$-2x_2 = -3 \quad \therefore x_2 = \frac{3}{2}$$

$$x_1 + x_2 = 2 \quad \therefore x_1 = 2 - \frac{3}{2} = \frac{1}{2}$$

$\frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$

So Linear combn $= \frac{1}{2}(1,2,-1) + \frac{3}{2}(1,0,1) = (2,1,1)$

↳ means b lies exactly in the plane spanned by the 2 vect, that's why its proj is itself

(18) $P^2 = P$, P projects onto $C(A)$

$(I-P)^2 = I-P$ $\quad (I-P)$ " " $N(A^T)$

↓

both proj Mat.

$(I-P)^2 = I^2 - 2PI + P^2 = I - 2P + P = I - P$

if $P = A(A^TA)^{-1}A^T$

Then P projects any vect $b \in R^n$ onto $C(A)$.

$Pb \in C(A)$

Now, $e = b - Pb = \underline{(I-P)b}$

error vect, part of b ortho to A

$e \perp C(A)$

$\therefore e \in N(A^T)$

(10) A vect $v(x, y)$ such that,

$(x, y) \cdot (1, 1) = 0$

$\therefore x + y = 0$

x = -y (any vector of this form would do.

20

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ChatGPT

Step 1: Find two vectors in the plane $x - y - 2z = 0$

To project onto the plane, we need two linearly independent vectors that lie in the plane. These vectors form a basis for the column space of the matrix A, which will be used to compute the projection matrix P.

Equation of the plane:

$$x - y - 2z = 0$$

Let's find two vectors that lie in this plane.

Vector 1: Let $z = 0$

- If $z = 0$, the equation becomes $x - y = 0$, so $x = y$.
  Choose $x = 1$, then $y = 1$.
  So, one vector in the plane is $\mathbf{v_1} = (1, 1, 0)$.

Vector 2: Let $y = 0$

- If $y = 0$, the equation becomes $x - 2z = 0$, so $x = 2z$.
  Choose $z = 1$, then $x = 2$.
  So, another vector in the plane is $\mathbf{v_2} = (2, 0, 1)$.

These two vectors $\mathbf{v_1} = (1, 1, 0)$ and $\mathbf{v_2} = (2, 0, 1)$ are linearly independent and lie in the plane.

Step 2: Construct matrix A

Now, we'll construct the matrix A using these vectors as columns.
So:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Step 3: Compute $P = A(A^T$ A^T

Ask anything

Now we will compute the projection matrix P.

Step 3.1: Compute $A^T A$

$$A^T = \begin{vmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \end{vmatrix}$$

A^T A = \begin{bmatrix} 1 & 1 & 0 \ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \ 1 & 0 \ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \ 2 & 5 \end{bmatrix}]

Step 3.2: Compute $(A^T A)^{-1}$

Now, we compute the inverse of $A^T A$:

$$(A^T A) = \begin{vmatrix} 2 & 2 \\ 2 & 5 \end{vmatrix}$$

The determinant of $A^T A$ is:

$$\det(A^T A) = (2)(5) - (2)(2) = 10 - 4 = 6$$

The inverse of $A^T A$ is:

$$(A^T A)^{-1} = \frac{1}{6} \begin{vmatrix} 5 & -2 \\ -2 & 2 \end{vmatrix} = \begin{vmatrix} \frac{5}{6} & \frac{-1}{3} \\ \frac{-1}{3} & \frac{1}{3} \end{vmatrix}$$

Step 3.3: Compute $A(A^T A)^{-1} A^T$

Now, compute the projection matrix P:

$$P = A \begin{vmatrix} \frac{5}{6} & \frac{-1}{3} \\ \frac{-1}{3} & \frac{1}{3} \end{vmatrix} A^T$$

Final Result:

After the matrix multiplication, we get the projection matrix P:

$$P = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} & \frac{-1}{3} \\ \frac{1}{3} & \frac{-1}{3} & \frac{1}{3} \end{bmatrix}$$

This is the desired projection matrix onto the plane $x - y - 2z = 0$.

Ask anything

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Step 1: Find a vector $\mathbf{e}$ perpendicular to the plane

The normal vector to the plane $x - y - 2z = 0$ is $\mathbf{e} = (1, -1, -2)$. This vector is perpendicular to the plane because it points in the direction of the gradient of the plane's equation.

So, the vector $\mathbf{e}$ is:

$$\mathbf{e} = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$$

Step 2: Compute $Q = \frac{\mathbf{e}\mathbf{e}^T}{\mathbf{e}^T\mathbf{e}}$

We now calculate the projection matrix Q onto the line defined by $\mathbf{e}$. The formula for Q is:

$$Q = \frac{\mathbf{e}\mathbf{e}^T}{\mathbf{e}^T\mathbf{e}}$$

Compute $\mathbf{e}^T\mathbf{e}$:

$$\mathbf{e}^T\mathbf{e} = (1)^2 + (-1)^2 + (-2)^2 = 1 + 1 + 4 = 6$$

Compute $\mathbf{e}\mathbf{e}^T$:

$$\mathbf{e}\mathbf{e}^T = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & -1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ -2 & 2 & 4 \end{bmatrix}$$

Compute Q:

$$Q = \frac{1}{6} \begin{bmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ -2 & 2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{6} & \frac{-1}{6} & \frac{-1}{3} \\ \frac{-1}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{-1}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Thus, the matrix Q is:

$$Q = \begin{bmatrix} \frac{1}{6} & \frac{-1}{6} & \frac{-1}{3} \\ \frac{-1}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{-1}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Ask anything

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Step 3: Compute $P = I - Q$

The projection matrix P onto the plane is given by:

$$P = I - Q$$

Compute $I - Q$:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now subtract Q from I:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{6} & \frac{-1}{6} & \frac{-1}{3} \\ \frac{-1}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{-1}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Perform the subtraction:

$$P = \begin{bmatrix} 1 - \frac{1}{6} & 0 - \frac{-1}{6} & 0 - \frac{-1}{3} \\ 0 - \frac{-1}{6} & 1 - \frac{1}{6} & 0 - \frac{1}{3} \\ 0 - \frac{-1}{3} & 0 - \frac{1}{3} & 1 - \frac{2}{3} \end{bmatrix}$$

$$P = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} & \frac{-1}{3} \\ \frac{1}{3} & \frac{-1}{3} & \frac{1}{3} \end{bmatrix}$$

Thus, the projection matrix P onto the plane $x - y - 2z = 0$ is:

$$P = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} & \frac{-1}{3} \\ \frac{1}{3} & \frac{-1}{3} & \frac{1}{3} \end{bmatrix}$$

Conclusion:

The projection matrix P onto the plane $x - y - 2z = 0$ is:

$$P = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} & \frac{-1}{3} \\ \frac{1}{3} & \frac{-1}{3} & \frac{1}{3} \end{bmatrix}$$

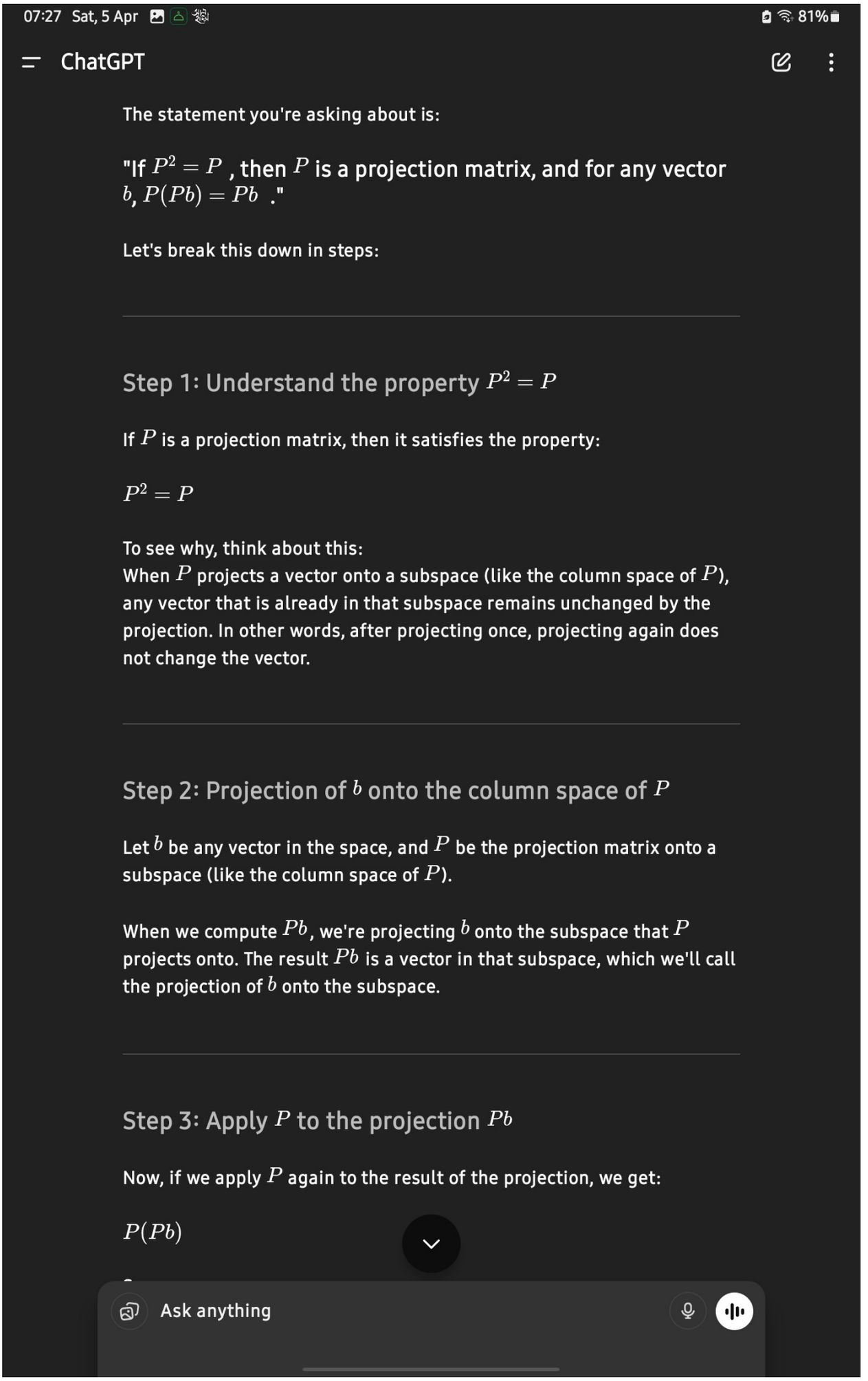
Ask anything

(22) $P = \left(A (A^TA)^{-1} A^T\right)^T = AA (A^TA)^{-1} (A^TA)^{-1} A^TA^T$

$= A \; A^TA (A^TA)^{-1} (A^TA)^{-1} A^T$

Pb is in Col space (where it project).

So P(Pb) is also Pb.



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Share a link to chat?
This creates a copy that others can chat with

not change the vector.

Step 2: Projection of b onto the column space of P

Let b be any vector in the space, and P be the projection matrix onto a subspace (like the column space of P).

When we compute Pb, we're projecting b onto the subspace that P projects onto. The result Pb is a vector in that subspace, which we'll call the projection of b onto the subspace.

Step 3: Apply P to the projection Pb

Now, if we apply P again to the result of the projection, we get:

$$P(Pb)$$

So:

$$P(Pb) = Pb$$

Conclusion

In summary:

- P is a projection matrix if $P^2 = P$.
- When P projects a vector b onto its column space, the result is Pb, which lies in the column space of P.
- Applying P again to Pb does not change it, i.e., $P(Pb) = Pb$.

This demonstrates that once b is projected onto the subspace, further projections of Pb will not alter the result.

Ask anything

(23) $I - \left(A(A^TA)^{-1}A^T\right)^T = I - A\left((A^TA)^{-1}\right)^TA$

$= A(A^TA)^{-1}A^T = A^T(\quad)$

$=$

the inverse of a symmetric matrix is symmetric.

So $((A^TA)^{-1})^T = (A^TA)^{-1}$ unchanged

(24) A → invertible · C(A) → all of R^n

So $P = I$ $P - I = 0$ $\therefore e = 0$

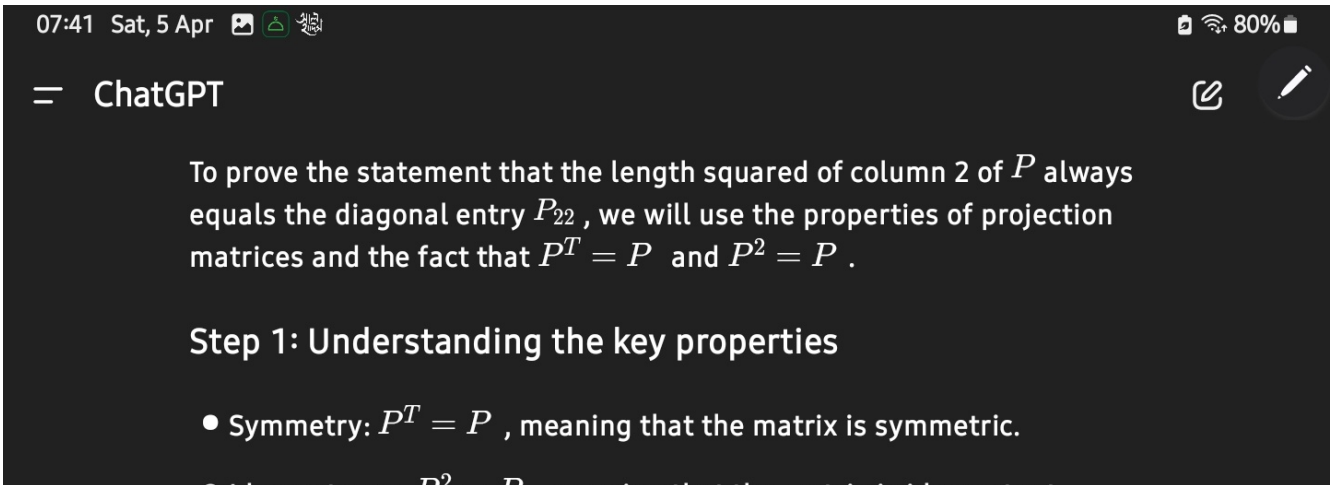
(25)

(27) mxm Mat A → rank m. → A^{-1} exists

$A^2 = A$

$A^2A^{-1} = AA^{-1}$ $\therefore A = I$ ✓

(29)



- Idempotence: $P^2 = P$, meaning that the matrix is idempotent (applying the projection twice is the same as applying it once).

We will prove the result by focusing on the (2,2) entry of P^2.

Step 2: (2,2) Entry of P^2

Let's look at the (2,2) entry of P^2, which is the dot product of row 2 of P with column 2 of P.

$$P^2 = P \cdot P$$

The (2,2) entry of P^2 is:

$$P_{22} = (\text{row 2 of } P) \cdot (\text{column 2 of } P)$$

Since P is symmetric ($P^T = P$), the row 2 of P is the same as the column 2 of P. So, the (2,2) entry of P^2 is just the squared length of row 2 (or column 2) of P.

Thus:

$$P_{22} = \|\text{row 2 of } P\|^2 = \|\text{column 2 of } P\|^2$$

This shows that the length squared of column 2 of P equals the diagonal entry P_{22}.

Step 3: Example

Let's now use a specific example to demonstrate this.

Consider the matrix:

$$P = \frac{1}{2}\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$$

Ask anything

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This matrix is symmetric, and we want to verify that the length squared of column 2 equals P_{22}.

- Step 3.1: Compute P^2

First, we compute P^2:

$$P^2 = \frac{1}{2}\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \cdot \frac{1}{2}\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$$

P^2 = \frac{1}{4} \begin{bmatrix} 2 & 2 \ 2 & 2 \end{bmatrix} = \frac{1}

{2} \begin{bmatrix} 1 & 1 \ 1 & 1 \end{bmatrix} = P] So, $P^2 = P$, which confirms that P is a projection matrix.

- Step 3.2: Compute the diagonal entry P_{22}

Now, look at the (2,2) entry of P:

$P_{22} = \frac{1}{2}$

- Step 3.3: Compute the length squared of column 2 of P

Column 2 of P is:

$$\text{column 2 of } P = \begin{vmatrix} 1/2 \\ 1/2 \end{vmatrix}$$

$$\|\text{column 2 of } P\|^2 = \left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}$$

- Step 3.4: Conclusion

We see that the length squared of column 2 of P is $\frac{1}{2}$, which is exactly equal to the diagonal entry P_{22} .

Final Conclusion

We have demonstrated that for the projection matrix P, the length squared of any column k equals th[illegible]agonal entry P_{kk} of P. In this specific example, the length squa[illegible] column 2 equals $P_{22} = \frac{1}{2}$. This property holds because of the symmetry and idempotence of projection

Ask anything

(30 After solve of 7 look above.